CO1-AT1-Case Study Analysis Report

Problem Statement: Smart Museum Artifact Preservation Platform

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1. Understand the Problem Statement

- The Smart Museum Artifact Preservation Platform is designed to monitor and protect valuable museum artifacts using digital technologies.
- Objectives: Monitor artifact conditions, detect environmental changes, predict preservation risks, manage digital archives, generate conservation reports, and notify museum authorities.
- Scope: Artifact monitoring, environmental sensing, risk prediction, digital archiving, reporting, and alert management.

2. Identify Key Stakeholders and Challenges

- Stakeholders: Museum Authorities, Curators, Conservation Experts, Visitors, IT Administrators, Archivists.
- Roles: Authorities manage operations; curators oversee collections; conservation experts preserve artifacts; IT teams maintain the system; archivists manage records.
- Challenges: Environmental damage, manual monitoring, data loss, delayed maintenance, security risks, and preservation cost.

3. Analyze the Current Approach

- Current Process: Manual inspections, periodic environmental checks, paper-based or basic digital records.
- Strengths: Human expertise and established conservation practices.
- Limitations: Time-consuming, error-prone, delayed detection of risks, and limited predictive capabilities.

- Gaps: Lack of real-time monitoring, automated alerts, and centralized preservation data.

4. Proposed Software Solution Using Software Engineering Principles

- Modules: Artifact Monitoring, Environmental Monitoring, Risk Prediction, Digital Archive Management, Report Generation, Notification System.
- Functional Requirements: Track artifact condition, collect sensor data, predict risks, store archives, generate reports, send alerts.
- Non-Functional Requirements: Security, reliability, scalability, maintainability, usability, and performance.
- Software Engineering Principles: Modular design, scalable architecture, reliable monitoring, secure data storage, and user-friendly interfaces.

5. Justify the Chosen Methodology/Framework

- Selected SDLC Model: Agile Scrum.
- Justification: Supports incremental development, continuous feedback, rapid updates, and easier adaptation to museum requirements.
- Benefits: Improves collaboration, reduces project risk, and simplifies maintenance.

6. Expected Outcomes and Limitations

- Expected Outcomes: Improved artifact preservation, reduced environmental damage, enhanced security, better documentation, and higher operational efficiency.
- How Challenges Are Addressed: Real-time monitoring, predictive analytics, centralized archives, and automated notifications.
- Limitations: Initial setup cost, sensor dependency, staff training requirements, and maintenance effort.
- Future Enhancements: AI-based image analysis, IoT expansion, cloud integration, and advanced predictive models.

Code of Conduct:

I V.Rakshana-192425137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Rakshana